

CO2-AT2-CSA0911

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

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CO2-AT2-Assignment

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QUESTIONS**Q1** Bank Token Management System
10 marks**Q2** Employee ID Normalization System
10 marks**Q3** Student Email Username Extractor
10 marks

2/3 answered

Q1 Bank Token Management System 10 marksCustomers receive tokens in the order they arrive at a bank counter.
Requirements:
Add customer tokens to a Queue
Serve customers in FIFO order
Display the next customer to be served and the remaining queue**Your Code**

```
1 import java.util.LinkedList;
2 import java.util.Queue;
3 import java.util.Scanner;
4
5 public class BankTokenSystem {
6     public static void main(String[] args) {
7         Queue<String> tokenQueue = new LinkedList<>();
8         Scanner scanner = new Scanner(System.in);
9
10        while (scanner.hasNextLine()) {
11            String input = scanner.nextLine();
12
13            if (input.isEmpty()) {
14                continue;
15            }
16        }
17    }
18 }
```

InputToken1
Token2
Token3
display**Output**Token added: Token1
Current queue: [Token1]
Token added: Token2
Current queue: [Token1, Token2]
Token added: Token3
Current queue: [Token1, Token2, Token3]

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QUESTIONS**Q1** Bank Token Management System
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Q2 Employee ID Normalization System 10 marksAn HR department receives employee IDs in mixed formats such as emp1023. Create a Java program to:
Remove spaces using trim()
Convert the alphabetic prefix to uppercase using toUpperCase()
Join the prefix and number using concat()**Your Code**

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         if (scanner.hasNextLine()) {
8             String input = scanner.nextLine().trim();
9
10            int splitIndex = -1;
11            for (int i = 0; i < input.length(); i++) {
12                if (Character.isDigit(input.charAt(i))) {
13                    splitIndex = i;
14                    break;
15                }
16            }
17        }
18    }
19 }
```

Input

emp1023

Output

EMP1023

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QUESTIONS

Q1 Bank Token Management System
10 marks**Q2** Employee ID Normalization System
10 marks**Q3** Student Email Username Extractor
10 marks

2/3 answered

Q3 Student Email Username Extractor

10 marks

A college portal stores student emails such as arun23@college.edu. Develop a program that:

Finds the position of @ using indexOf()
Extracts the username using substring()
Displays the username length using length()

Your Code

```
1 import java.util.Scanner;  
2  
3 public class Main {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6  
7         if (sc.hasNextLine()) {  
8             String email = sc.nextLine();  
9             int atPosition = email.indexOf('@');  
10  
11             if (atPosition != -1) {  
12                 String username = email.substring(0, atPosition);  
13                 System.out.println(username);  
14                 System.out.println(username.length());  
15             }  
16         }  
17     }  
18 }
```

Input

arun23@college.edu

Output

arun23
6